

08 Geometric Transformations



8.5 Orthogonal Projection

Projection matrices via tangent or normal vectors — symmetric, rank-one, idempotent, singular — from the plane to 3D space

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Build a planar projection matrix from a tangent vector τ :
 $P = \tau \tau^T$

3

Tangent and normal vectors are mutually orthogonal; $[\hat{n}, \hat{\tau}]$ forms an orthonormal basis

5

A projection matrix has determinant 0 and is singular — projection loses information

7

A planar projection matrix has rank one; 3D and higher-dimensional ones do not

2

Build 2D, 3D, and higher-dimensional projection matrices from a normal vector n : $P = I - \hat{n} \hat{n}^T$

4

A projection matrix is symmetric: $P = P^T$

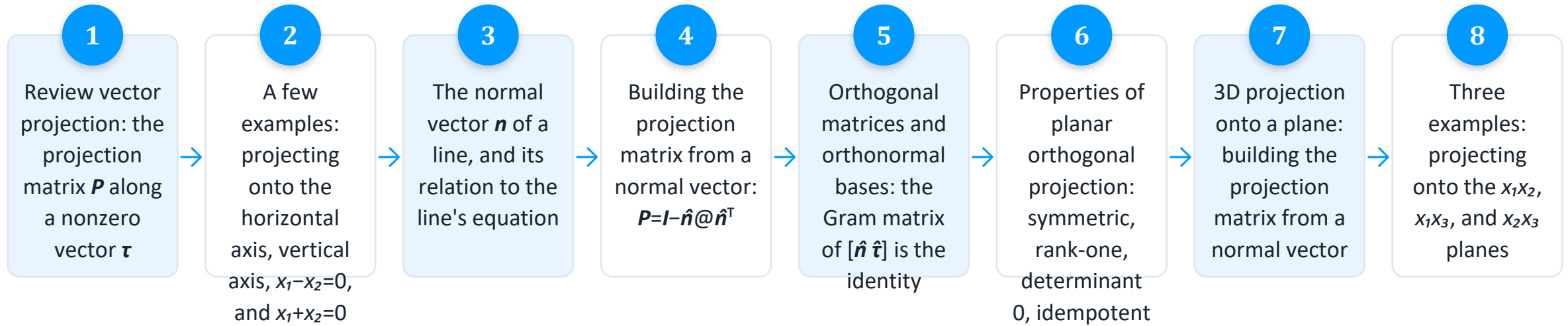
6

Repeated projection leaves the result unchanged: $P @ P = P$, i.e. projection matrices are idempotent

8

In 3D, build projection matrices onto coordinate planes using normal vectors

From Tangent and Normal Vectors to 3D Projection



Seven Lenses for Every Transform in This Chapter

- 1 Matrix-multiplication view: how the matrix acts on $\mathbf{0}$, $\mathbf{e}_1$, $\mathbf{e}_2$, and the all-ones vector $\mathbf{1}$
- 2 The determinant: $|\det|$ greater than 1? Positive, negative, or zero? Determinants of special matrices
- 3 How the determinant reflects the area or volume change caused by the transform
- 4 The inverse (does it exist?) and the inverse transform; inverses of special matrices
- 5 Linear-combination view: parallelogram/parallelepiped grids, and whether dimension drops
- 6 Does the transform matrix form a basis, and what kind of basis?
- 7 Properties of the transform matrix: symmetric, diagonal, orthogonal, invertible, commutative

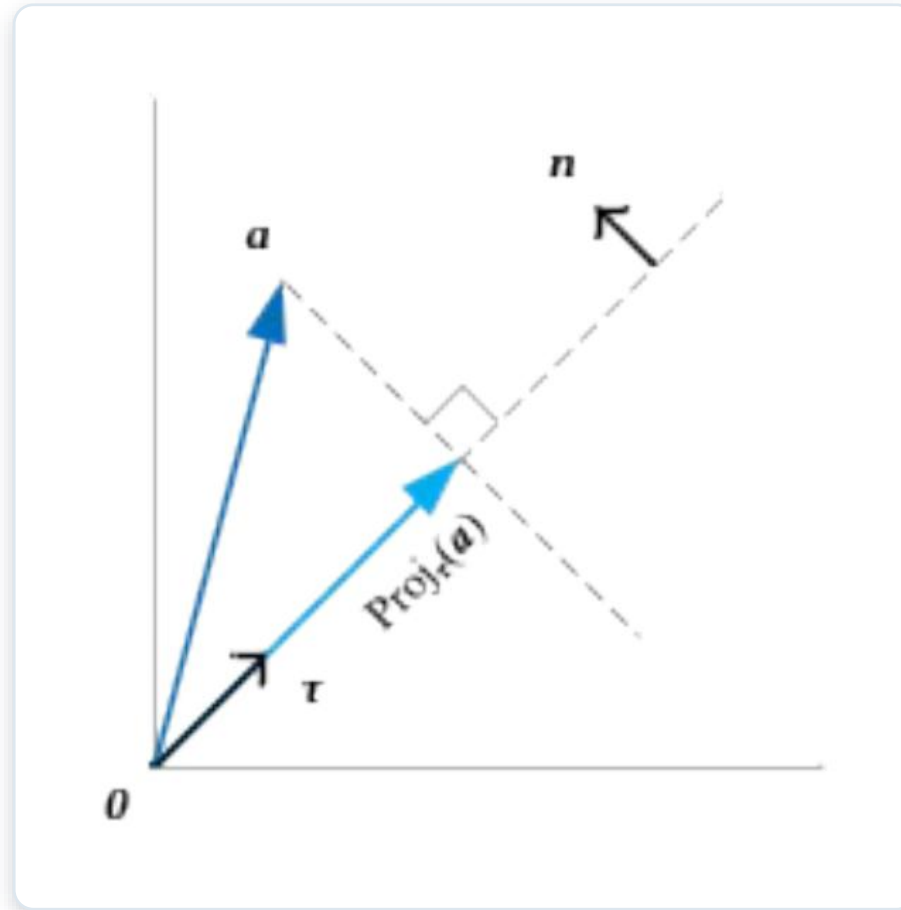
Projecting onto a nonzero vector τ : the line through the origin with tangent τ gives projection matrix P

$$P = \tau @ \tau^T / (\tau^T @ \tau) = \hat{\tau} @ \hat{\tau}^T$$

- A line's tangent vector describes which way the line points; writing τ as a unit direction vector $\hat{\tau}$ makes this clearly a rank-one matrix
- A rank-one matrix is the outer product of a nonzero column vector and a nonzero row vector — all its rows (or columns) are linearly dependent

Figure 1 · Tangent Vector of a Line Through the Origin

- Vector $\mathbf{a}$ projects onto $\boldsymbol{\tau}$ to give $\text{Proj}_{\boldsymbol{\tau}}(\mathbf{a})$; $\mathbf{n}$ is perpendicular to $\boldsymbol{\tau}$ — tangent and normal vectors always come in pairs



Projecting $\mathbf{a}$ onto $\boldsymbol{\tau}$: the scalar-projection and inner-product forms agree

$$\text{proj}_{\boldsymbol{\tau}}(\mathbf{a}) = \|\mathbf{a}\| \cos\theta \times \hat{\boldsymbol{\tau}} = (\mathbf{a} \cdot \boldsymbol{\tau} / \boldsymbol{\tau} \cdot \boldsymbol{\tau}) \boldsymbol{\tau}$$

- This book covered vector projection earlier: scalar projection times the direction vector equals the inner-product form
- This projection formula is the foundation for building the projection matrix $\mathbf{P}$

Projection matrices for projecting onto the horizontal axis ($x_2=0$) and vertical axis ($x_1=0$)

$$P(\text{horizontal}) = [1, 0; 0, 0] \quad P(\text{vertical}) = [0, 0; 0, 1]$$

- The horizontal axis has tangent $\tau=[1,0]^T$; after projection x_1 is unchanged and x_2 becomes 0
- The vertical axis has tangent $\tau=[0,1]^T$; after projection x_1 becomes 0 and x_2 is unchanged

Figure 2 · Projecting onto the Horizontal and Vertical Axes

- (a) Projecting onto the horizontal axis: x_1 unchanged, x_2 becomes 0; (b) onto the vertical axis: x_1 becomes 0, x_2 unchanged

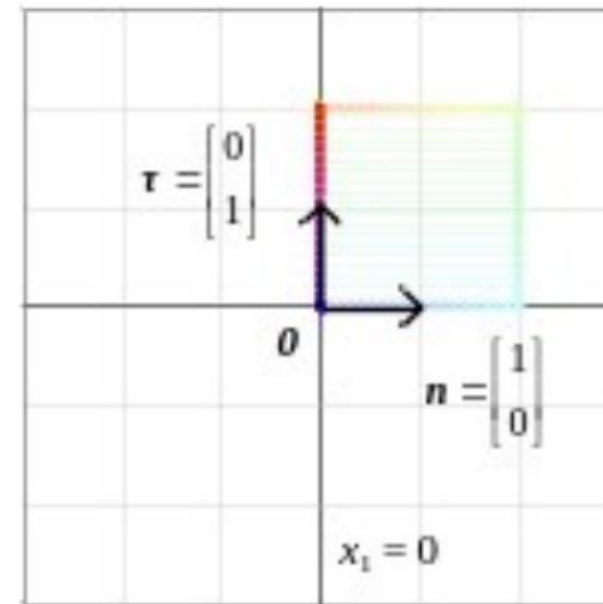
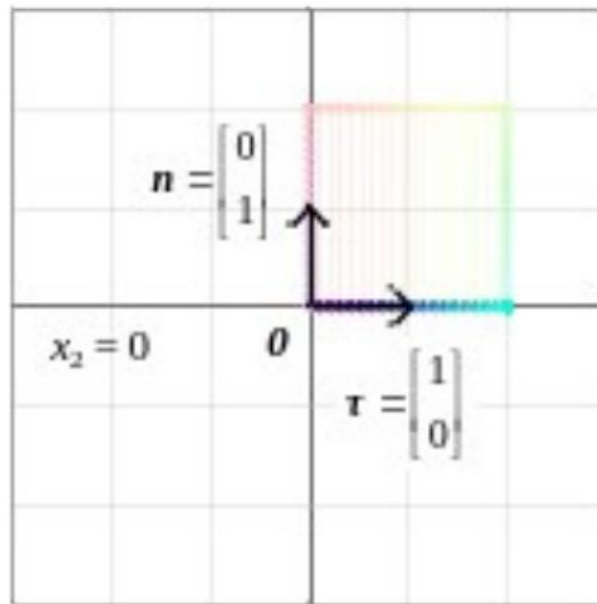


图 2. 向横轴、纵轴投影

Projection matrices for $x_1 - x_2 = 0$ and $x_1 + x_2 = 0$

$$P(x_1 - x_2 = 0) = [0.5, 0.5; 0.5, 0.5] \quad P(x_1 + x_2 = 0) = [0.5, -0.5; -0.5, 0.5]$$

- $x_1 - x_2 = 0$ has tangent $\tau = [1, 1]^T$; after normalizing, the projected coordinates are equal — points land on this line
- $x_1 + x_2 = 0$ has tangent $\tau = [1, -1]^T$; after projection the two coordinates are opposite in sign — points land on this line

Figure 3 · Projecting onto $x_1 - x_2 = 0$ and $x_1 + x_2 = 0$

- (a)(b) show the tangent τ and normal n of the two diagonal lines through the origin

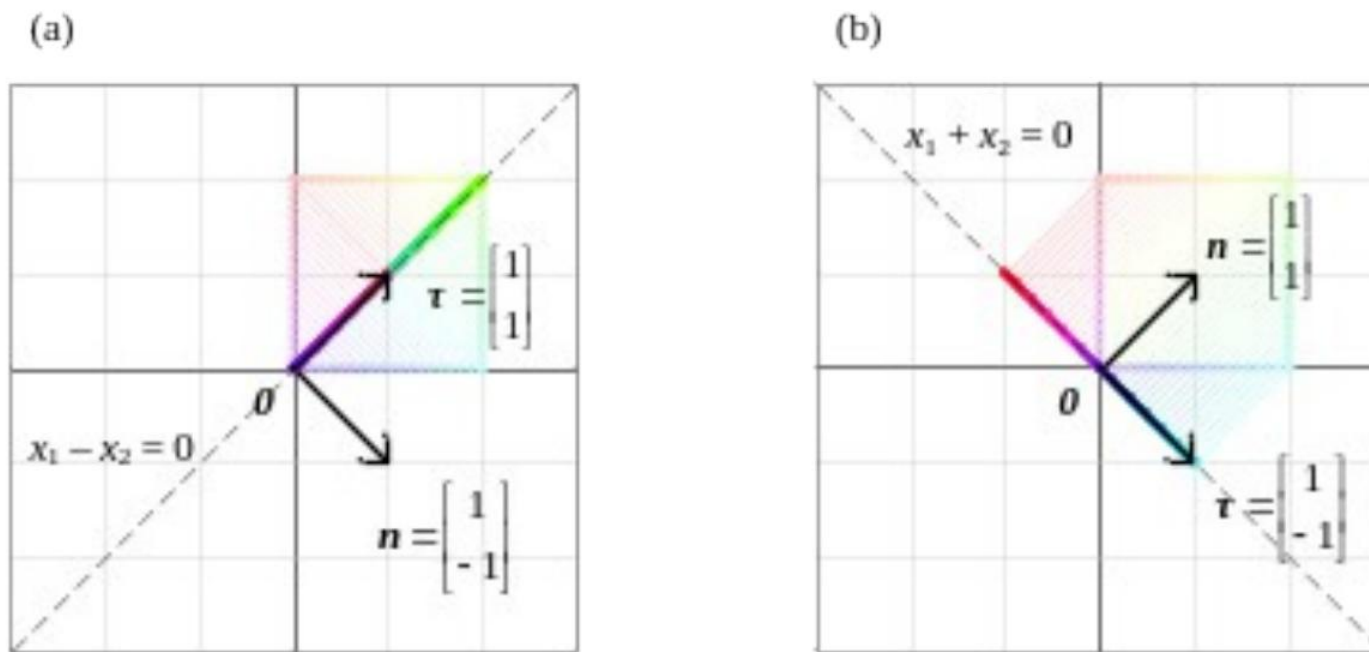


图 3. 向 $x_1 - x_2 = 0$ 、 $x_1 + x_2 = 0$ 投影

The normal vector n of the line $a_1x_1+a_2x_2=b$

$$n = [a_1, a_2]^T$$

- A line's normal vector is perpendicular to the line; a plane's normal vector is perpendicular to the plane
- Note: a normal vector is usually a free vector — it need not start at the origin

Figure 4 · Unit Normal Vectors of Lines in the Plane

- (a)(b) The unit normals of vertical and horizontal lines point along x_1 and x_2 ; (c) the normal of $x_1 + x_2 = \text{constant}$ is $[\sqrt{2}/2, \sqrt{2}/2]^T$

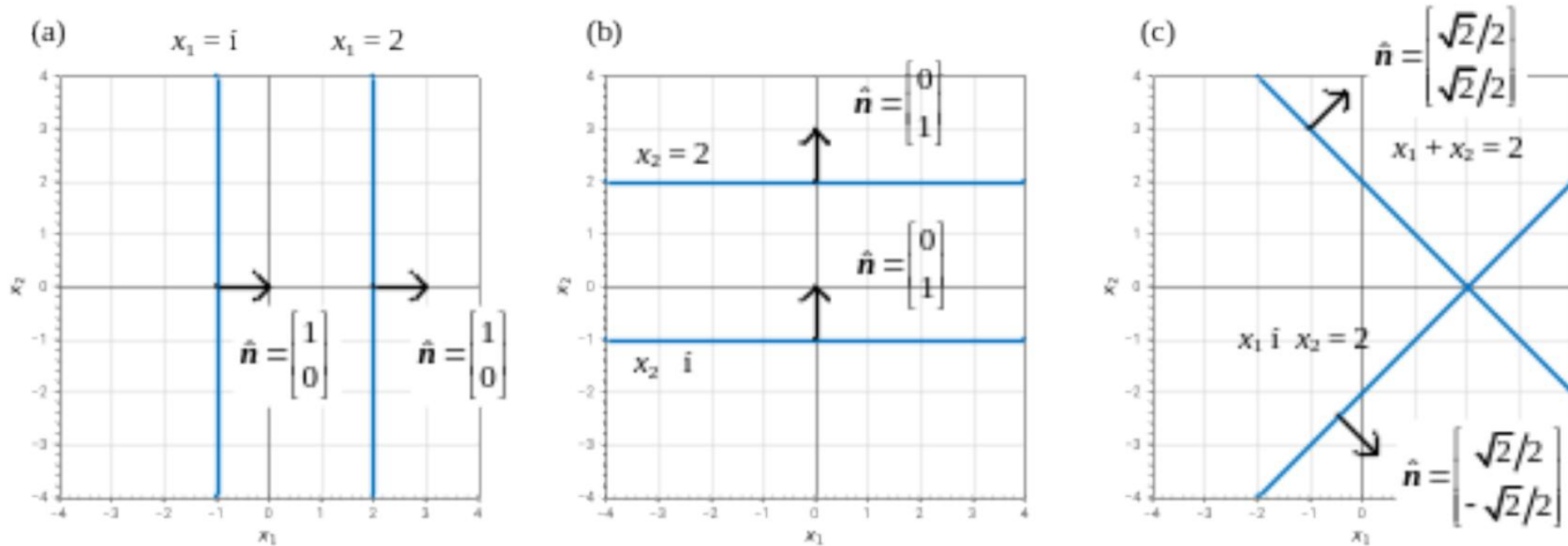


图 4. 平面直线单位法向量

Project $\mathbf{a}$ onto $\mathbf{n}$ (normal), then subtract to get $\mathbf{a}$'s orthogonal projection onto τ

$$\mathbf{a} - \text{proj}_{\mathbf{n}}(\mathbf{a}) = (\mathbf{I} - \mathbf{n}\mathbf{n}^T / (\mathbf{n}^T\mathbf{n})) \mathbf{a}$$

- By orthogonal decomposition, projecting $\mathbf{a}$ onto $\mathbf{n}$ and subtracting gives $\mathbf{a}$'s orthogonal projection along the tangent τ
- This gives the projection matrix onto τ : $\mathbf{P} = \mathbf{I} - \mathbf{n}\mathbf{n}^T / (\mathbf{n}^T\mathbf{n})$

Figure 5 · Normal Vector of a Line Through the Origin

- Subtracting $\mathbf{a}$'s projection onto $\mathbf{n}$ from $\mathbf{a}$ gives its orthogonal projection along τ

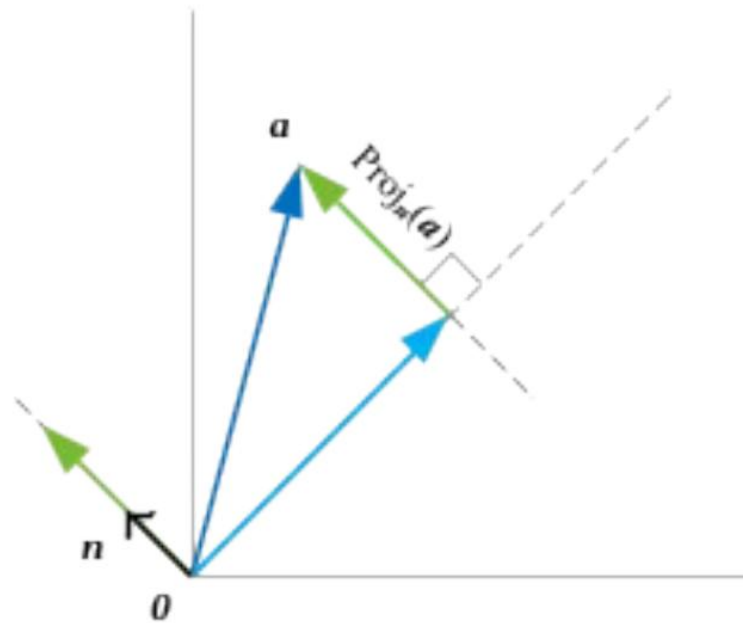


图 5. 过原点直线法向量

Substituting $n=[a_1,a_2]$ into $P=I-\hat{n}@\hat{n}$ expands to

$$P = (1 / (a_1^2 + a_2^2)) [a_2^2, -a_1 a_2; -a_1 a_2, a_1^2]$$

- For higher-dimensional orthogonal projection, this normal-vector construction is the usual choice
- A different derivation: with $\mathbf{n}=[a_1,a_2]^T$, $\mathbf{\tau}=[-a_2,a_1]^T$, using $\mathbf{\tau}@\mathbf{\tau}^T/(\mathbf{\tau}^T@\mathbf{\tau})$ gives the same result

Figure 6 · Two Orthogonal Projection Examples, via Normal Vectors

- Try computing the projection matrices for both lines shown, using tangent vectors and normal vectors separately

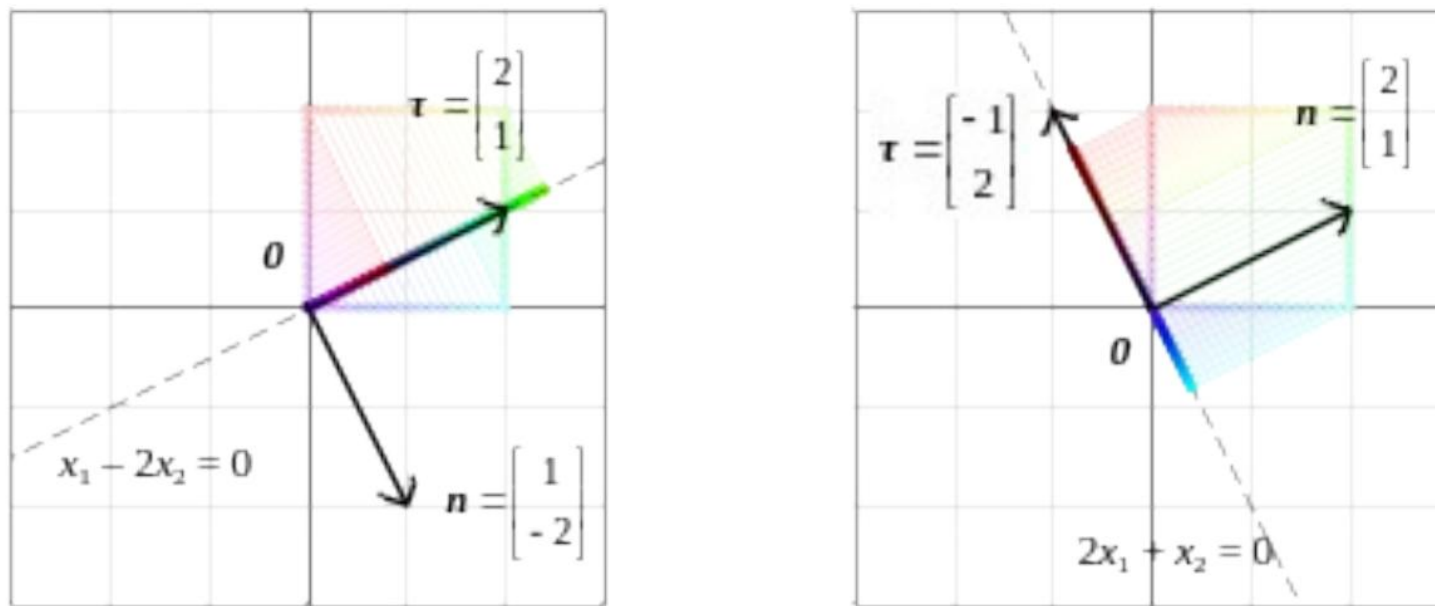


图 6. 两个正交投影变换的例子，法向量

The unit normal and tangent $[\hat{n} \ \hat{\tau}]$ form an orthonormal basis

$$\hat{n}\hat{n}^T + \hat{\tau}\hat{\tau}^T = I \quad [\hat{n} \ \hat{\tau}][\hat{n} \ \hat{\tau}]^T = I$$

- The two projection matrices sum to I , so the Gram matrix of $[\hat{n} \ \hat{\tau}]$ is the identity
- This means $[\hat{n} \ \hat{\tau}]$ is an orthogonal matrix — an important result we will rely on in the next chapter

The projection matrix P is symmetric, and it is also rank one

$$P = P^T$$

- The columns of P are linearly dependent, consistent with expansions like $P = a_2[a_2, -a_1] - a_1[a_2, -a_1]$
- Note: P being rank one holds only for planar orthogonal projection — 3D and higher-dimensional projection matrices are not rank one

A planar orthogonal projection matrix always has determinant 0

$$\det(\mathbf{P}) = 0$$

- This means $\mathbf{P}$ compresses a planar shape and information is lost
- A determinant of 0 means the projection matrix is singular — the key feature distinguishing projection from rotation and scaling

Figure 7 · Orthogonal Projection Is Not Invertible

- The original square is compressed to a line segment — the original shape cannot be recovered from the projected result

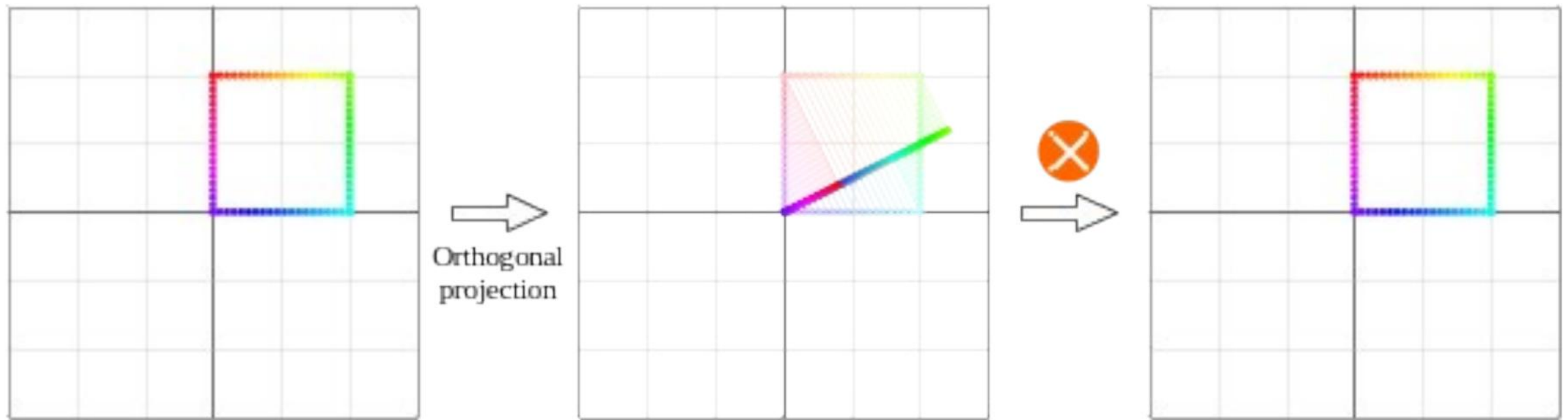


图 7. 正交投影变换不可逆

A projection matrix satisfies $P@P=P$, so it is called idempotent

$$P@P = P$$

- An idempotent matrix, multiplied by itself, still equals itself
- Geometrically, projecting a point repeatedly gives exactly the same result as projecting once

Figure 8 · Repeated Orthogonal Projection Leaves the Result Unchanged

- Applying orthogonal projection twice to the same shape: the second projection changes nothing further

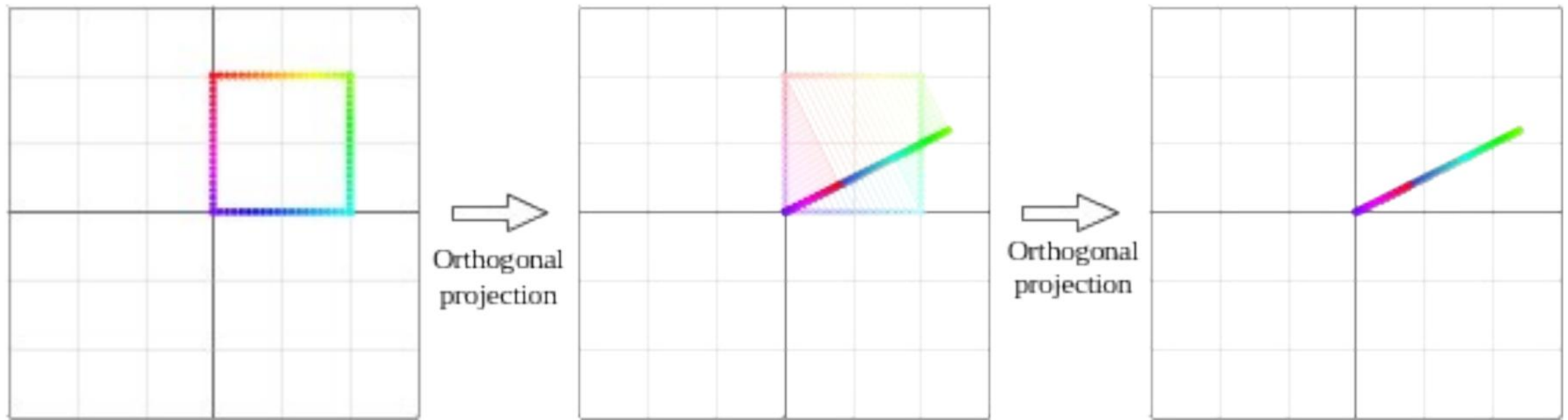


图 8. 连续正交投影结果不变

In 3D and higher dimensions, the normal-vector construction has the advantage

$$P = I - \hat{n} @ \hat{n}^T$$

- For a plane, its normal vector is simply the vector perpendicular to it
- This formula holds in any dimension — all you need is the unit normal $\hat{n}$ of the hyperplane

Figure 9 · Normal Vectors of Planes, in 3D Space

- (a) The x_1x_2 plane has normal $[0,0,1]^T$; (b) the x_1x_3 plane has normal $[0,1,0]^T$; (c) the x_2x_3 plane has normal $[1,0,0]^T$

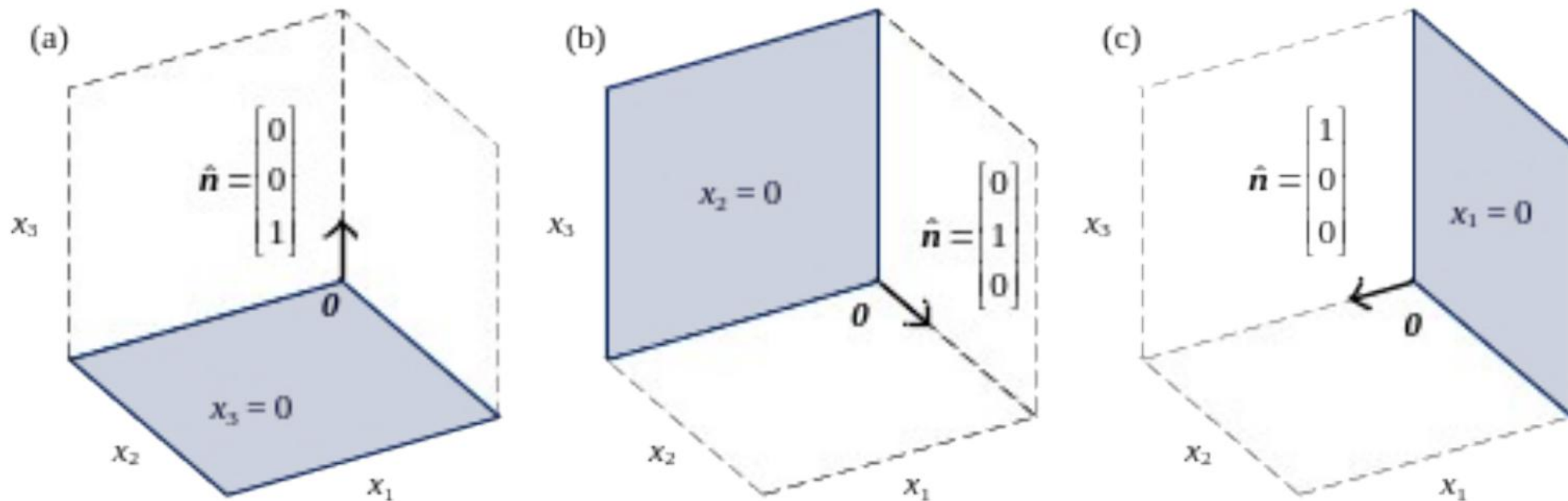


图 9. 平面的法向量

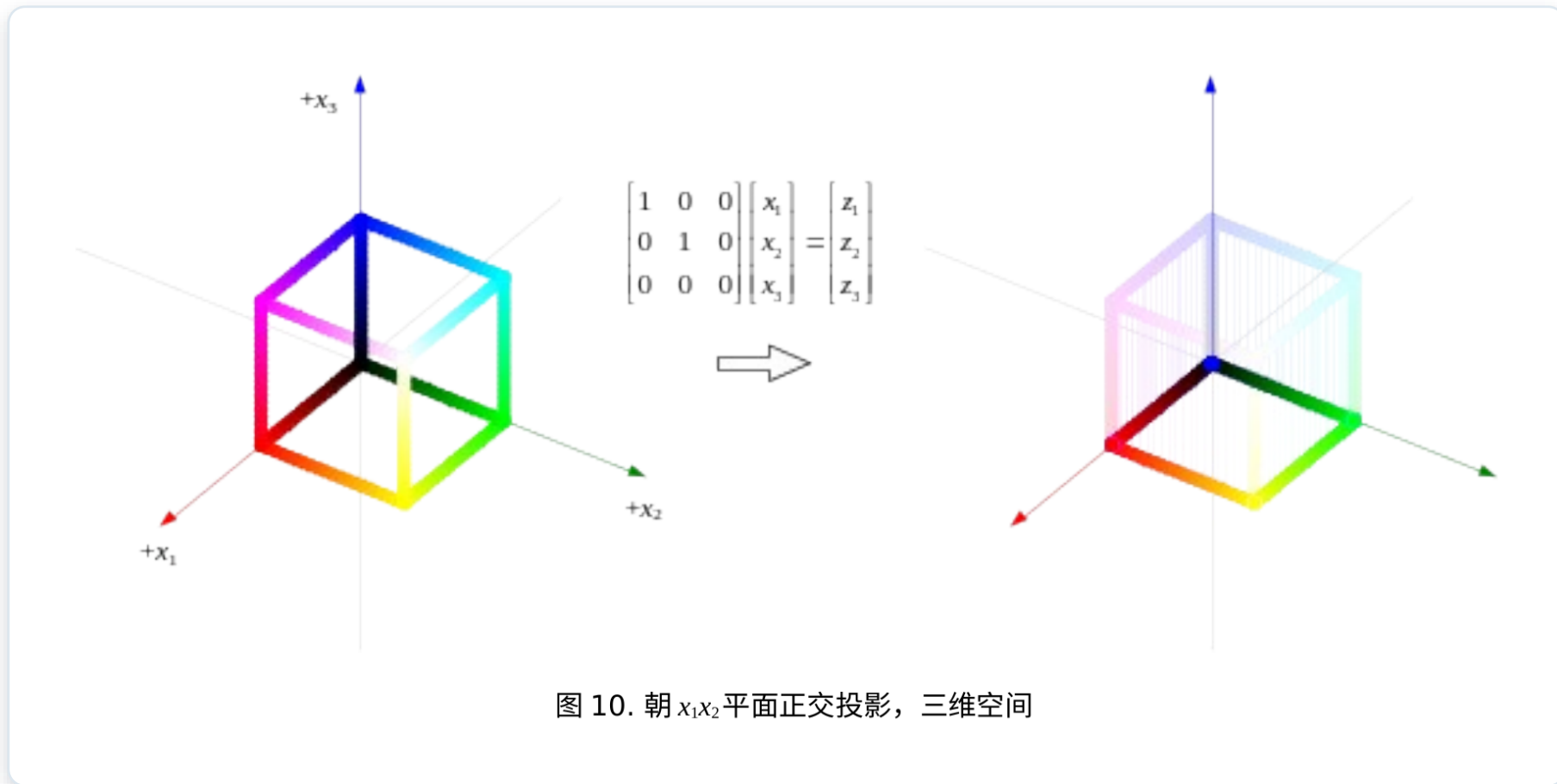
The unit normal of the x_1x_2 plane is $[0,0,1]$; its projection matrix is

$$P = \text{diag}(1, 1, 0)$$

- After projecting a 3D column vector $\mathbf{x}$ onto the x_1x_2 plane, x_1 and x_2 are unchanged and x_3 becomes 0
- Try computing this projection matrix's rank and determinant — is it still rank one?

Figure 10 · Orthogonal Projection onto the x_1x_2 Plane, in 3D

- The unit cube is flattened onto a square in the x_1x_2 plane — all information along x_3 is lost



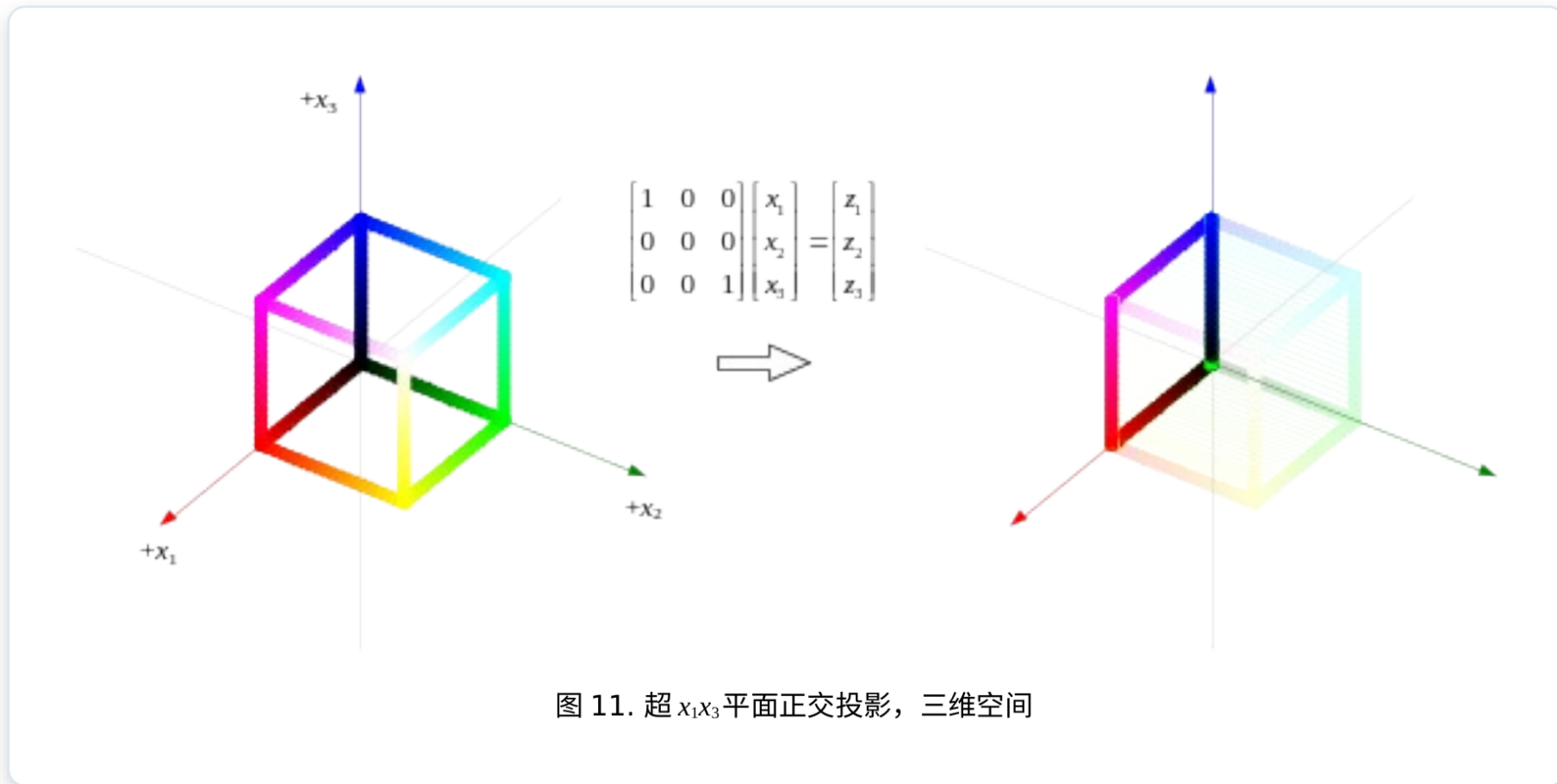
The unit normal of the x_1x_3 plane is $[0,1,0]$; its projection matrix is

$$P = \text{diag}(1, 0, 1)$$

- After projecting a 3D column vector $\mathbf{x}$ onto the x_1x_3 plane, x_1 and x_3 are unchanged and x_2 becomes 0

Figure 11 · Orthogonal Projection onto the x_1x_3 Plane, in 3D

- The unit cube is flattened onto a square in the x_1x_3 plane — all information along x_2 is lost



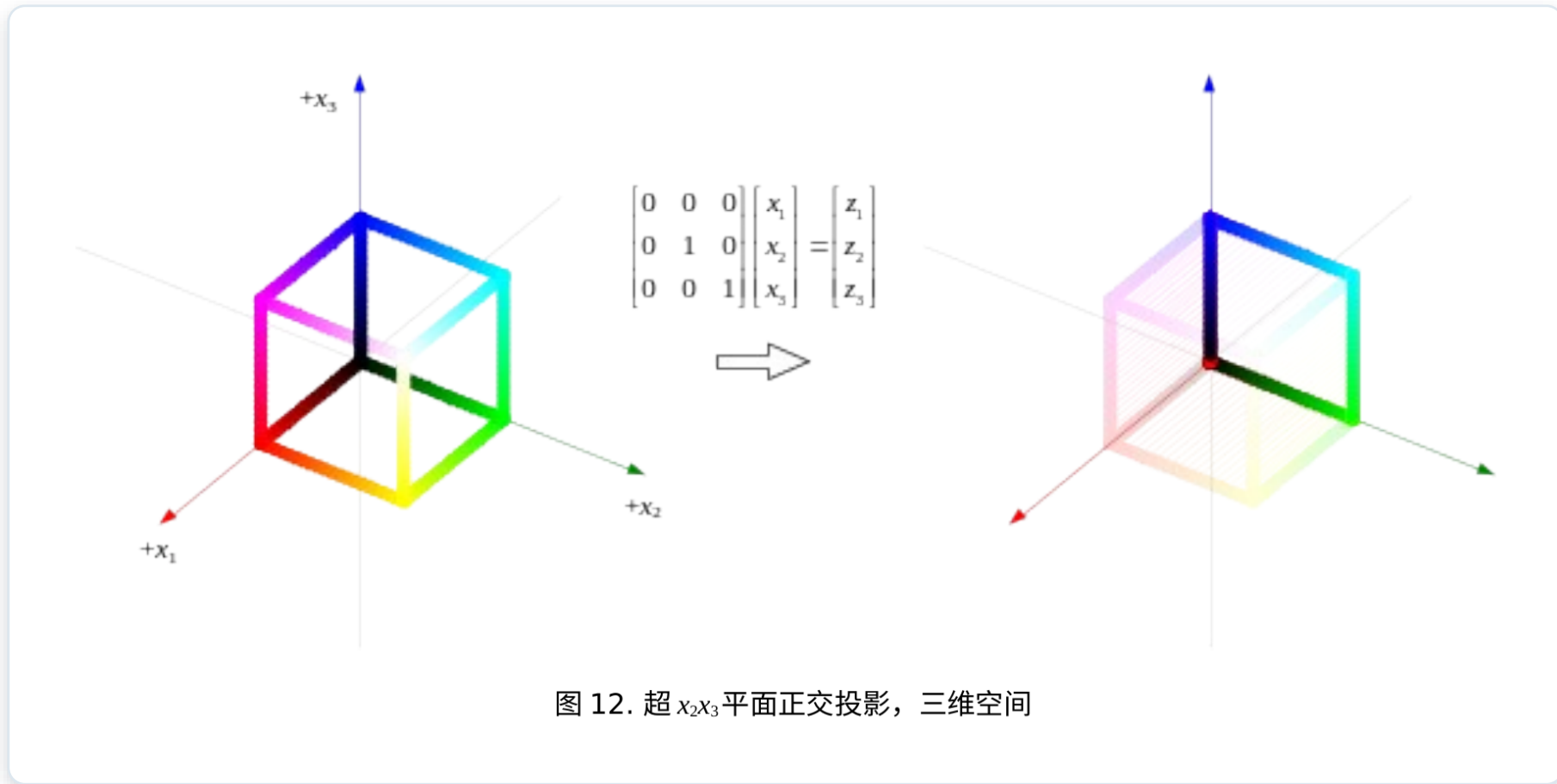
Try computing the orthogonal projection matrix onto the x_2x_3 plane

$$P = \text{diag}(0, 1, 1)$$

- The unit normal of the x_2x_3 plane is $[1,0,0]$; after projection x_1 becomes 0 while x_2, x_3 are unchanged
- Where does the information lost in projection go? How could it be recovered? These are questions for the next chapter

Figure 12 · Orthogonal Projection onto the x_2x_3 Plane, in 3D

- The unit cube is flattened onto a square in the x_2x_3 plane — all information along x_1 is lost



SUMMARY

What Have We Seen from the Matrix View of Projection?

Tangent-vector construction $\mathbf{P}=\hat{\mathbf{t}}\hat{\mathbf{t}}^T$ and normal-vector construction $\mathbf{P}=\mathbf{I}-\hat{\mathbf{n}}\hat{\mathbf{n}}^T$ reach the same place

A projection matrix $\mathbf{P}$ is symmetric and idempotent ($\mathbf{P}\mathbf{P}=\mathbf{P}$), with determinant always 0 — it is singular

A planar projection matrix has rank one; 3D and higher-dimensional ones do not

The unit normal and tangent form an orthonormal basis $[\hat{\mathbf{n}} \ \hat{\mathbf{t}}]$ — a key building block for the next chapter

Glossary

Orthogonal Projection 正交投影 — projecting a vector onto a line or plane along the perpendicular direction

Normal Vector 法向量 — n , the vector perpendicular to a line or plane

Rank-One Matrix 秩一矩阵 — a matrix expressible as the outer product of a column and a row vector

Orthonormal Basis 规范正交基 — formed by unit normal and tangent vectors, e.g. $[\hat{n} \ \hat{t}]$

Symmetric Matrix 对称矩阵 — a matrix satisfying $P=P^T$

Tangent Vector 切向量 — τ , the vector describing a line's direction

Projection Matrix 投影矩阵 — P , the matrix that performs orthogonal projection

Idempotent Matrix 幂等矩阵 — a matrix satisfying $P@P=P$

Gram Matrix 格拉姆矩阵 — $[\hat{n} \ \hat{t}]^T[\hat{n} \ \hat{t}]$, the identity for an orthonormal basis

Hyperplane 超平面 — the higher-dimensional generalization of a plane, defined via a normal vector

EXPLORE FURTHER

Further Reading

Recap · 8.3 Rotation

Earlier in this chapter — rotation matrices, orthogonal matrices, and Gram matrices

Further Reading · 8.6 Reflection

Later in this chapter — reflection transforms, another case of orthogonal matrices

Further Reading · 9.1 Orthogonality

Next chapter — orthogonal vectors, orthogonal matrices, and orthonormal bases in full

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Q1 – Q9

Q1 For lines through the origin such as $-x_2=0$, $-x_1=0$, $x_1-2x_2=0$, compute the tangent vector, normal vector, and projection matrix (two ways).

Q3 Compute the orthogonal projections of point $(3, 4)$ onto the unit vectors $[0,1]^T$, $[1,0]^T$, $[3/5,4/5]^T$, $[4/5,-3/5]^T$.

Q5 Modify LA_08_02_01.ipynb to visualize the planar orthogonal projections in Figures 2 and 3.

Q7 For hyperplane normal vectors $[0,0,0,1]^T$, $[0,0,1,0]^T$, $[1,1,1,1]^T$, compute the projection matrices.

Q9 For projecting onto a line in the plane that does not pass through the origin, think through how to compute the projected point.

Q2 For 3D planes through the origin such as $x_2=0$, $x_1=0$, $x_1+x_2=0$, compute the normal vector and projection matrix.

Q4 A line through the origin makes a 30° angle with the horizontal axis. Compute the orthogonal projection of point $(3, 4)$ onto it.

Q6 Modify LA_08_03_01.ipynb to visualize the 3D orthogonal projections in Figures 10, 11, and 12.

Q8 For hyperplanes $x_1+x_2+x_3+x_4=0$ and $x_1-x_2-x_3+x_4=0$, compute the projection matrices.

Q10 – Q17

Q10 Write a Python function `proj_matrix_tangent(tau)` that returns the projection matrix for a given tangent vector.

Q12 Use numpy to verify: for any nonzero vector v , the projection matrix P satisfies $P@P$ equals P .

Q14 Compute the rank of any planar projection matrix P from this section and verify with `numpy.linalg.matrix_rank`.

Q16 Use matplotlib to visualize an arbitrary vector a and its projection $\text{proj}(a)$ onto the line $x_1+2x_2=0$.

Q11 Write a Python function `proj_matrix_normal(n)` that returns the projection matrix for a given normal vector.

Q13 Use numpy to verify: every projection matrix P from this section satisfies P equals P transposed.

Q15 Write code comparing: projecting point $(2, 3)$ onto $x_1-x_2=0$ once, then projecting the result again — are the two results the same?

Q17 Given the plane normal vector $n=[1,2,2]/3$, verify $\|n\|$ equals 1 and compute the corresponding projection matrix.

Q18 – Q25

Q18 Write a Python function `is_projection_matrix(P)` that checks whether a matrix is both symmetric and idempotent.

Q20 For the 3D plane through the origin with normal $n=[1,1,1]/\sqrt{3}$, compute the projection matrix and verify its rank is 2.

Q22 Think through and compute: is the identity matrix I minus a rank-one projection matrix still a projection matrix?

Q24 Explore: the eigenvalues of an orthogonal projection matrix can only be 0 or 1 — verify this with `numpy.linalg.eig` on this section's examples.

Q19 Explore: if the tangent vector τ is not normalized, does $P=\tau\tau^T/(\tau^T\tau)$ still give the correct projection matrix? Verify with code.

Q21 Use Python to compare projecting a batch of points onto one line via matrix multiplication versus point-by-point computation.

Q23 Comprehensive: for the 4D hyperplane $x_1+x_2+x_3+x_4=0$, compute the projection matrix in Python and verify idempotency.

Q25 Preview thought: the next section covers reflection — how does the reflection matrix relate to the orthogonal projection matrix? (Hint: $\text{reflection} = I - 2 \times \text{projection}$)

10 Prompts to Explore With AI

- 1 Explain why a projection matrix P can be constructed either from a tangent vector or a normal vector, and how the two relate.
- 2 Prove that a projection matrix P satisfies P equals P transposed, and also P times P equals P .
- 3 Explain why a planar orthogonal projection matrix's determinant is always 0, and what that implies about the information in a figure.
- 4 Using Python, demonstrate projecting a square onto several different lines and observe how the projected result changes.
- 5 Explain the relationship between the orthonormal basis $[\hat{n} \ \hat{t}]$ and orthogonal matrices, and why this result matters for the next chapter.

10 Prompts to Explore With AI

6 Explain why higher-dimensional orthogonal projection is more naturally constructed from a normal vector.

7 Use Python to verify that projections onto the 3D coordinate planes are always diagonal matrices.

8 Explain why the eigenvalues of an orthogonal projection matrix can only be 0 or 1, and what that means geometrically.

9 Explain the relationship between reflection and orthogonal projection — why is the reflection matrix I minus twice the projection matrix?

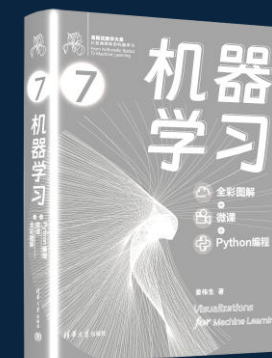
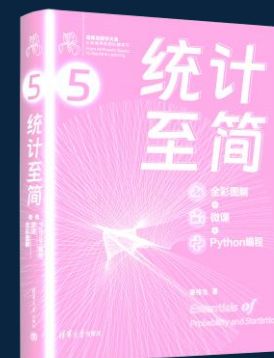
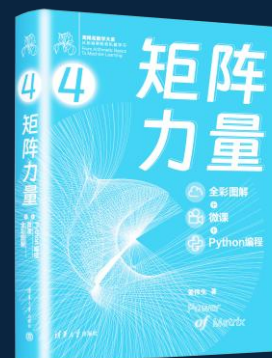
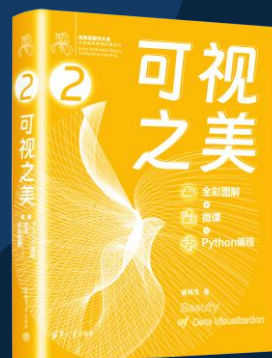
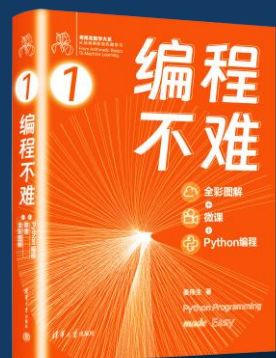
10 Explain what role orthogonal projection might play in machine learning and dimensionality reduction, such as PCA.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

